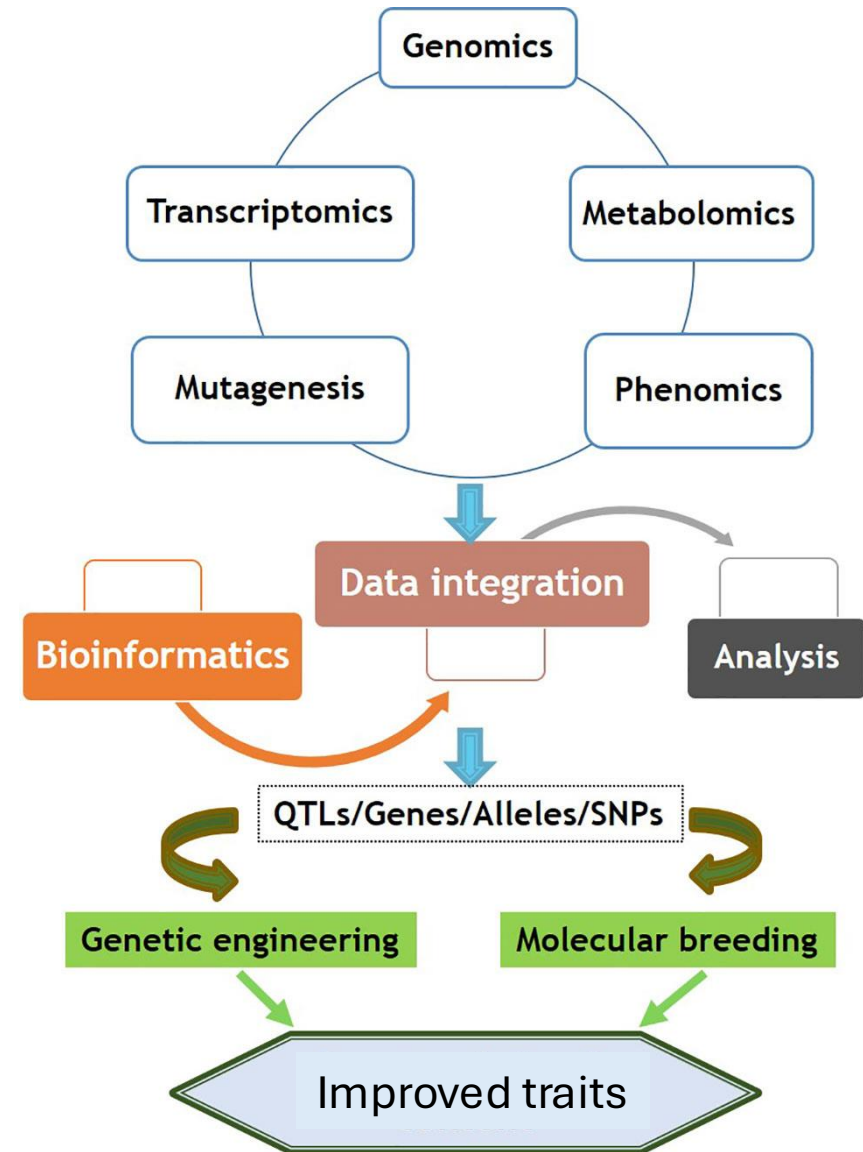
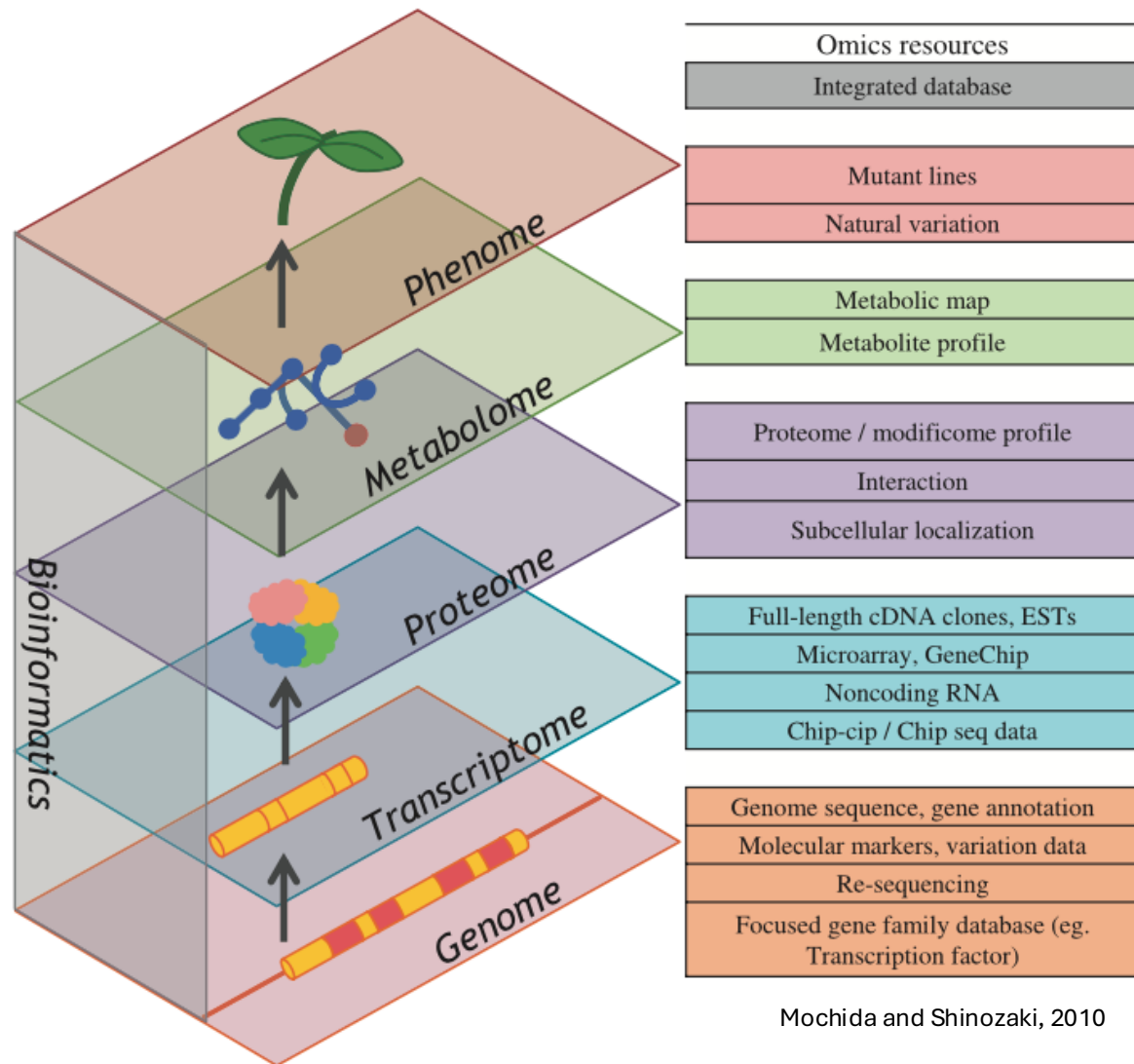




Coconut Genome Landscape

Siwaret Arikrit | Kasetsart University

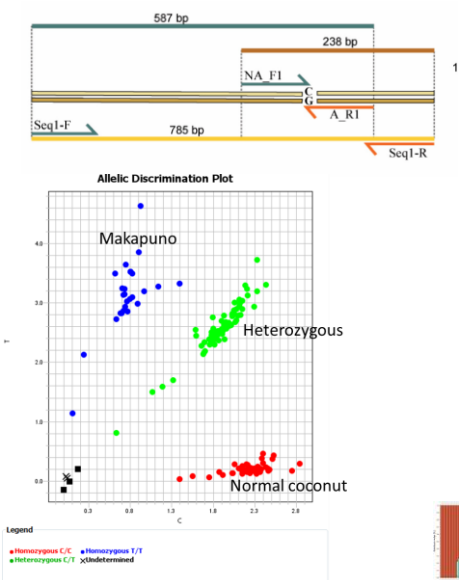
Omic spaces and related resources in plants



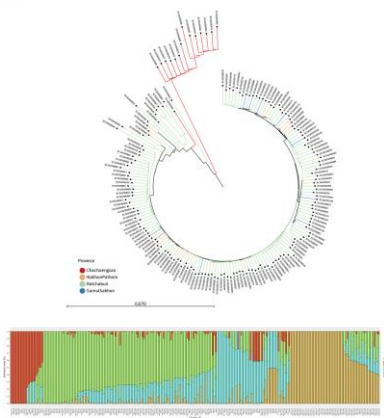
Application of multi-omics in coconut research

Genomics

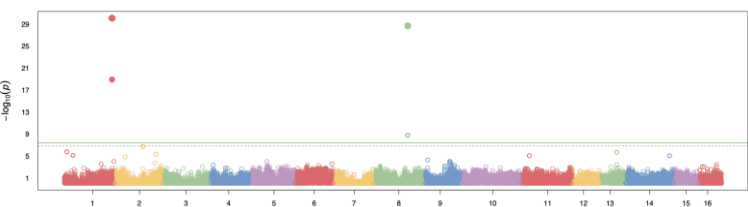
Marker development



Population study

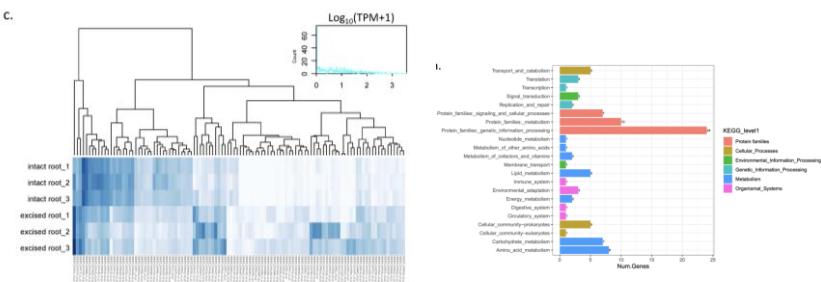


Gene/QTL identification

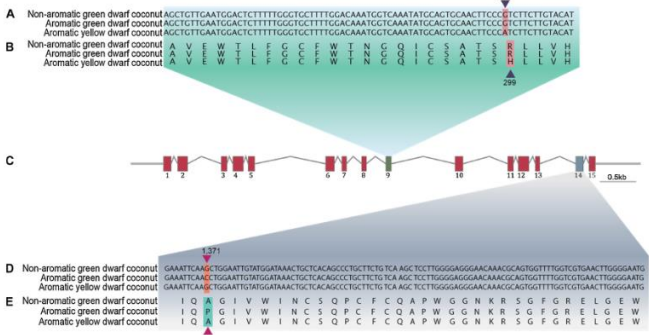


Transcriptomics

Gene expression analysis

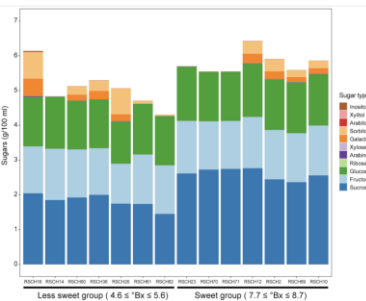
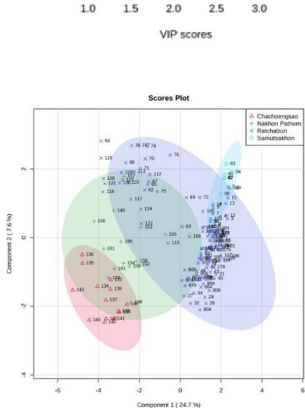
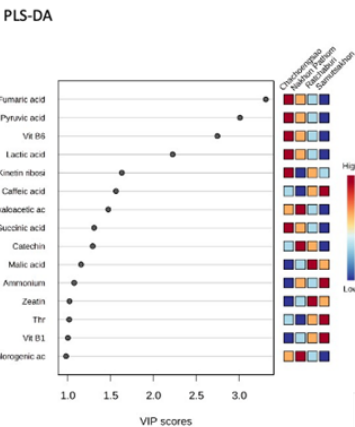


Orthologous gene identification

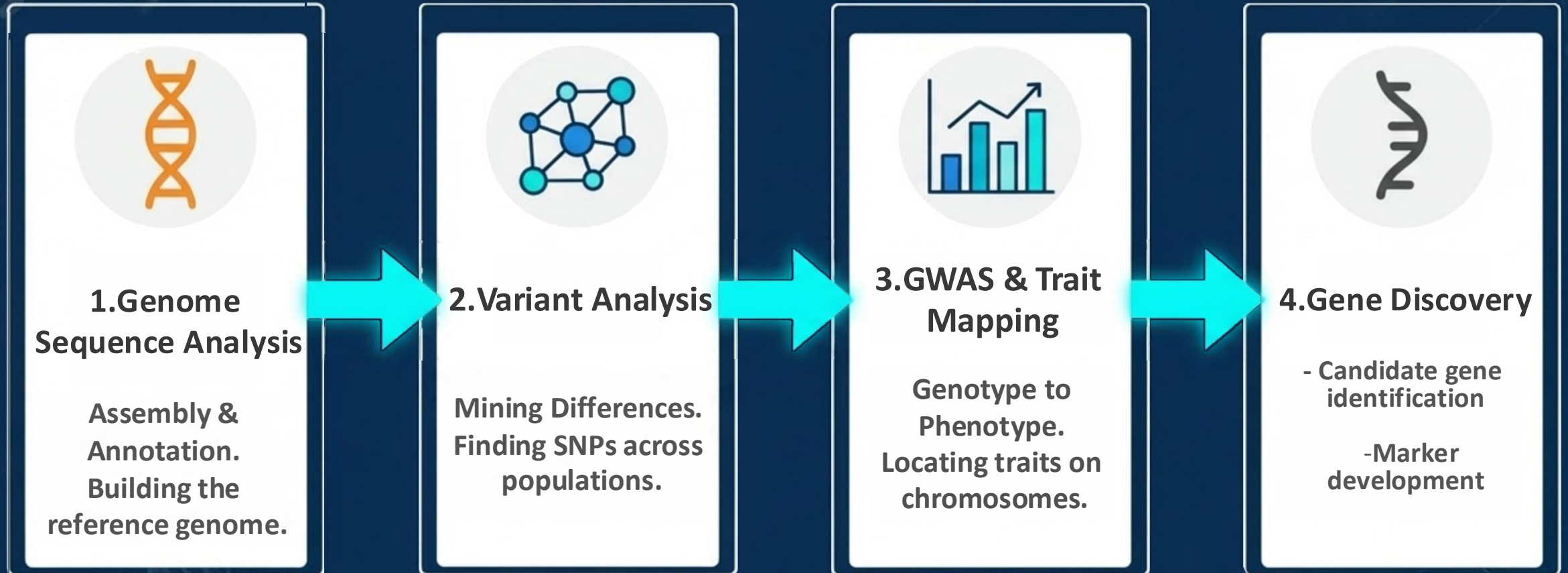


Metabolomics

Metabolic profiling



The Data-to-Discovery Workflow



Coconut Genome: Status of Reference Genomes

Genome Features

Coconut is diploid ($2n = 32$, 16 chromosomes) like many palms

Large proportion of transposable elements linked to genome size (>70% of repetitive content)

Genome size: approx. 2.4 Gb

Current Draft and complete genomes published

- *Hainan Tall (HT)*:

- Scaffold
(2017)

- Chromosome level
(2021)

- Chromosome level + Hi-C
(2023 - PAG)

- *Catigan Green Dwarf (CATD)*:

- Scaffold level
(2019)

DATA NOTE

The genome draft of coconut (*Cocos nucifera*)

Yong Xiao^{1,†}, Pengwei Xu^{3,†}, Haikuo Fan^{1,†}, Luc Baudouin^{4,5,†}, Wei Xia¹,
Stéphanie Bocs^{4,5}, Junyang Xu³, Qiong Li², Anping Guo², Lixia Zhou¹, Jing Li¹,

<https://doi.org/10.1038/s42003-020-01593-x>

OPEN

Coconut genome assembly enables evolutionary analysis of palms and highlights signaling pathways involved in salt tolerance

Yaodong Yang^{1,8}, Stéphanie Bocs^{2,3,4,8}, Haikuo Fan^{1,8}, Alix Armero^{3,8}, Luc Baudouin^{2,3,8},



De Novo Genome Sequence Assembly of Dwarf Coconut (*Cocos nucifera* L. 'Catigan Green Dwarf') Provides Insights into Genomic Variation Between Coconut Types and Related Palm Species

Darlon V. Lantican,^{*,†} Susan R. Strickler,[‡] Alma O. Canama,^{*} Roanne R. Gardoce,^{*} Lukas A. Mueller,[‡]
and Hayde F. Galvez^{*,§,1}

^{*}Genetics Laboratory, Institute of Plant Breeding, College of Agriculture and Food Science, University of the Philippines Los Baños, College, Laguna, Philippines 4031, [†]Philippine Genome Center, University of the Philippines System, Diliman,

Current Draft and complete genomes published

- *Chowghat Green Dwarf (CGD)*:

Assembly and Annotation of the Nuclear and Organellar Genomes of a Dwarf Coconut (Chowghat Green Dwarf) Possessing Enhanced Disease Resistance

Rajesh Krishna Muliya¹, Pallem Chowdappa¹, Santosh Kumar Behera², Sandeep Kasaragod²

- Scaffold level (2020)
- Chromosome level (2025)

OPEN

Chromosome scale genome assembly and annotation of coconut cultivar Chowghat Green Dwarf

M. K. Rajesh^{1,2,3}, Roli Budhwar³, Rohit Shukla³, Praveen Kumar Oraon³, Shailendra Goel⁴

- *Unknown tall (Cn. tall) and unknown dwarf (Cn. dwarf)*:

- Chromosome level (2021)

High-quality reference genome sequences of two coconut cultivars provide insights into evolution of monocot chromosomes and differentiation of fiber content and plant height



Shouchuang Wang^{1,2,3†}, Yong Xiao^{1,4†}, Zhi-Wei Zhou^{5,6†}, Jiaqing Yuan^{7†}, Hao Guo^{2†}, Zhuang Yang^{2,3}, Jun Yang²

- *MaWa (Malayan yellow dwarf x West African tall hybrid)*:

- Gold standard genome (Gap-free) (2025)

Gap-free, haplotype-resolved genome assembly reveals allele-specific expression and fruit color regulation in coconut

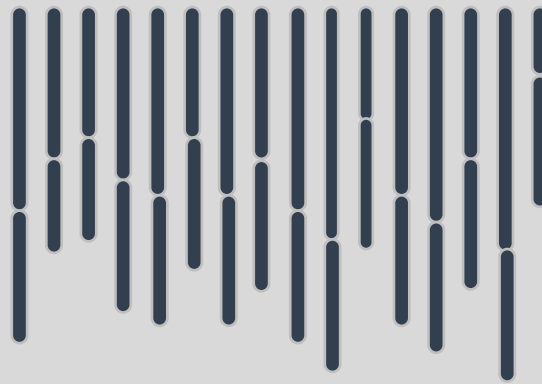
Min Chen^{a,b,1}, Jing Li^{a,c,1}, Ziqi Ye^{a,b}, Xiwei Sun^{a,c}, Xiaomei Liu^{a,c}, Hao Ding^{a,c}, Lixia Zhou^{a,c}

Thai Coconut genome

16 Chromosomes

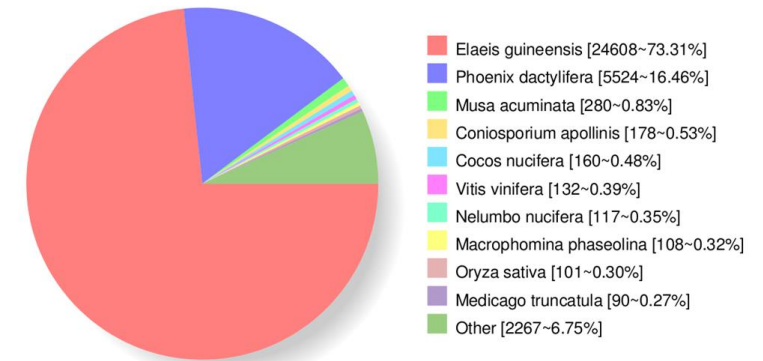
Estimated genome size 2.68 Gb

40,723 genes



	Thai coconut
Number of contigs	21,021
Total size of contigs (bp)	2,684,422,759
largest contig (bp)	15,219,968
Contig N50 count	1,253,973
Contig N90 count	45,496
GC content	37.82

Nr Homologous Species Distribution



Welcome to Coconut Genome Browser

The Coconut Genome Browser (CGB) is a platform for the most economically important coconut varieties. Data available from CGB includes gene annotation, gene expression, and genetic molecular markers. CGB also provides some useful genomic tools online.

Introduction

“

For hundreds of years, the coconut – fruit of the coconut palm (*Cocos nucifera* L.) – has been a great source of versatility. It provides food, drink, clothing and shelter, as well as income from its products. Due to its continuous production, the coconut is readily available all year round. Today, the main producers¹¹.

News

How to cite
[October, 2020]



- Genome browser (JBrowse)
- Expression profiles
- Blast search tools
- Gene search

Thai coconut reference



DNA

Genomics data

- Illumina sequencing data
 - 428.29 M reads
 - 128.48 Gb
- PACBIO
 - 160 Gb
- Illumina Hi-C
 - 200 Gb



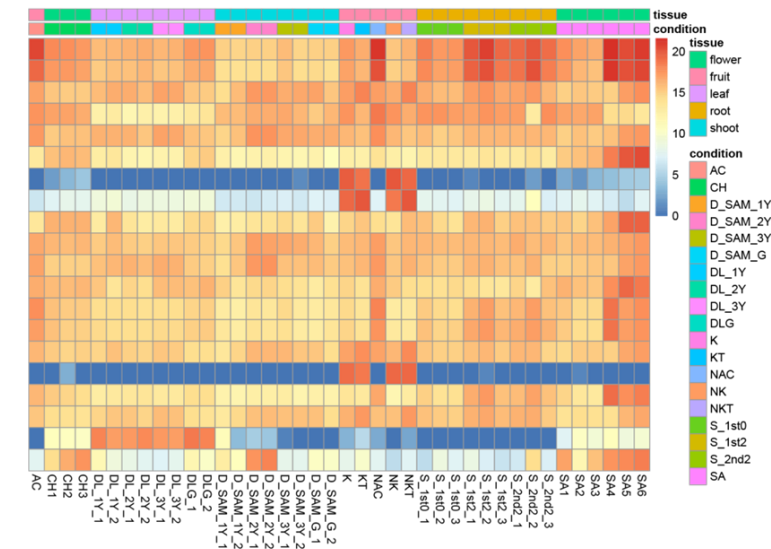
RNA

Transcriptomics data

- Root
 - 9 samples
- Inflorescent
 - 16 samples
- Fruit
 - 6 samples
- Shoot
 - 8 samples
- Leaf
 - 8 samples

Total data: 72.55 M reads (115.8 Gb)

~71,327 transcripts



Sequencing cost is dramatically decreased

